

$$= 3.36 \times 10^{19} \text{ J (max KE of e) [B1]}$$

7 (a)

As nuclear fission takes place very quickly and the number of neutrons emitted will be very large as the stages proceed on [1] (resulting in a chain reaction) and hence leading to a very large amount of energy in a short time, which may cause an explosion if not controlled [1]

7 (b)(i)

$$E = a^{n-1}b \rightarrow \lg E = (n-1)\lg(a) + \lg(b)$$

Thus the intercept of the graph is $\lg(b) - \lg(a)$ [M1]

$$\text{Gradient of graph, } \lg(a) = (4.6 - 2.7) / (6.8 - 2.0) = 0.3958$$

$$4.6 = (0.3958)(6.8) + c$$

$$c = 1.908 \quad [M1]$$

$$\text{therefore } \lg(b) - \lg(a) = 1.908$$

$$\lg(b) = 2.30$$

$$b = 200 \text{ MeV} \quad [A1]$$

7 (b)(ii)

Energy release due to one nuclear fission reaction of uranium

7 (c)(i)

$$92 - 36 = 56$$

7 (c)(ii)

The range of percentage yield is very large and hence in order to show the variation of the values in a single graph, a logarithmic scale is used.

7 (c)(iii)

$$\text{Percentage yield of having 135 and 99} = 6$$

$$\text{If the Uranium produces two nuclei with the same masses, mass of each nuclide} = 236/2 = 118$$

$$\text{Percentage yield of masses equal} = 0.015$$

$$\text{Therefore, the ratio} = 6/0.015$$

$$= 400$$

7 (d)(i)

Gamma (photon) radiation.

7 (d)(ii)

$$2477 \text{ TWh} = 2477 \times 10^{12} \times 3600$$

$$= 8.92 \times 10^{18} \text{ J} \quad [C1]$$

Since the efficiency is 35%,

The amount of heat required

$$= 100/35 \times 8.92 \times 10^{18} = 2.55 \times 10^{19} \text{ J} \quad [A1]$$

7 (d)(iii)

$$\begin{aligned} \text{Power required} &= (2.55 \times 10^{19}) / (336 \times 24 \times 60 \times 60) \\ &= 8.78 \times 10^{11} \text{ W} \quad [B1] \end{aligned}$$

Number of reactions needed in one second

$$\begin{aligned} &= (8.78 \times 10^{11}) / (167 \times 10^6 \times 1.6 \times 10^{-19}) \\ &= 3.28 \times 10^{22} \quad [B1] \end{aligned}$$

Amount required (mass per s)

$$\begin{aligned} &= 3.28 \times 10^{22} \times 235u \\ &= 3.28 \times 10^{22} \times 235(1.66 \times 10^{-27}) \\ &= 1.28 \times 10^{-2} \text{ kg s}^{-1} \quad [A1] \end{aligned}$$

7 (e)

The decay chain of Mo has a relatively longer half life and poses a health hazard if it is not stored properly over an extended period of time. And so it has to be securely shielded (or encapsulated) and stored in a safe and isolated place for a longer period of time.

The decay chain of Xe has much shorter half life and so the time it spends in isolated storage is much less and it can be 'disposed off' safely after a few days. However, due to its high activity at the initial stage, the shielding provided must be sufficiently 'thick' to contain its radiation.

7 (f)

- Like fossil fuel, there is finite amount of uranium available for nuclear reactor plants to meet the increasing needs for energy supply. They are not renewable and so when the supply drops, alternative energy sources will be required. Its usage is not sustainable in the long run.
- Despite raising the safety standards of nuclear reactor plants, it is still subjected to potential natural disaster (eg earthquake and tsunami). In the event of such disruption, nuclear meltdown can occur.
- Nuclear plants generate lots of heat due to the poor efficiency of heat engines. Its contribution to global warming is substantial and it can lead to climate changes.
- For certain countries where land is scarce (eg Singapore)), setting up a nuclear reactor plant would not be feasible as there needs to be a minimum safe distance between populated areas and the nuclear site in case of a nuclear meltdown.